ModelIt!

Teacher's Guide

Should We or Shouldn't We?

When Science Meets Hard Choices

Quick Reference

Grade Level	Grade NE
NGSS Standard	HS-ETS1-1, HS-ETS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ETS1-1, HS-ETS1-3

Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints.

Three-Dimensional Learning

Science Practice	**Engaging in Argument from Evidence** — Students use model outputs as evidence to argue for or against intervention in conservation scenarios.
Disciplinary Core Idea	**ETS1.B Developing Possible Solutions** — Students evaluate solutions by weighing criteria (problem severity, evidence strength) against constraints (solution risk, unintended consequences).
Crosscutting Concept	**Cause and Effect** — Students identify that three separate causes (problem severity, risk, evidence) combine to produce an effect (whether intervention is justified), and that predicting effects requires considering all inputs.

CAST Testing Alignment

MULTIPLE CHOICE:

Scientists propose releasing genetically modified mosquitoes to fight malaria. The disease kills 600,000 people/year, but the long-term ecological effects are unknown. Using your model framework, which factor creates the most uncertainty? A) Problem Severity B) Solution Risk C) Available Evidence D) Intervention cost

CONSTRUCTED RESPONSE:

A city proposes using AI-controlled drones to replant a burned forest. Using your model, evaluate this intervention by rating Problem Severity, Solution Risk, and Available Evidence. Then argue whether the city should proceed or wait for more research.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

As technology becomes more powerful, we face increasingly difficult decisions about when and how to use it. Environmental ethics is the branch of philosophy that helps us navigate these choices — especially when the stakes are high and the answers are not clear.

The Intervention Dilemma

When an ecosystem is in trouble, humans face a choice: intervene using technology, or let nature take its course. Both options have risks. Intervention might cause unintended consequences. But doing nothing might allow irreversible damage. There is no risk-free option — only different risks.

The Precautionary Principle

The precautionary principle states: when an action raises threats to the environment, precautionary measures should be taken even if some cause-and-effect relationships are not fully established scientifically. In other words, if something MIGHT be harmful, err on the side of caution. Critics argue this can prevent beneficial interventions.

Real-World Dilemmas

CRISPR gene drives could eliminate malaria-carrying mosquitoes but might cascade through food webs. Coral reef robots can plant coral fragments but may alter natural reef development. AI-managed wildlife reserves can detect poachers but raise surveillance concerns. Each dilemma involves the same three factors: severity, risk, and evidence.

Why Models Help

Ethical models do not provide answers — they provide STRUCTURE. They force decision-makers to explicitly rate each factor instead of relying on gut feelings. They reveal where disagreements actually lie (is it about the severity of the problem or the risk of the solution?). And they make the reasoning transparent so others can evaluate and challenge it.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Problem Severity, Solution Risk, and Available Evidence as external components (factors we can assess and rate) and Intervention Justified as the single internal component (the decision output).

E **Establish Relationships**
Students determine that Problem Severity and Available Evidence positively affect justification, while Solution Risk negatively affects it — three competing forces on one output.

V **Visualize & Model**
Students build a model with three external inputs (two positive, one negative) pointing to one internal output.

E **Evaluate Outcomes**
Students run clear-yes, clear-no, and hard-call scenarios to see how the balance shifts.

R **Revise & Extend**
Students add Cost, Public Support, or Reversibility to see how additional real-world constraints complicate the decision.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Imagine you have the technology to save a dying forest — but using it might create a NEW problem. Do you act?"*

Watch For

Present a real dilemma: Yellowstone controlled burns. Show photos of the 1988 fires and the recovery. Was the 'do nothing' approach right?

Slide 2: Learning Objectives

3 min

Say: *"Today we build a model for something you would not expect — ethical decision-making."*

Watch For

Pre-teach 'unintended consequences' with examples: cane toads in Australia, kudzu in the US South.

Slide 3: Big Question

3 min

Say: *"This is not a question with a right answer. It is a question about HOW to think clearly when the stakes are high."*

Watch For

Quick poll: 'A species is going extinct. A risky new technology might save it. Who says act? Who says wait?' Then ask: What would change your mind?

Slide 4: LEVER Framework

2 min

Say: *"We are going to model this decision the same way we model physical systems. Three inputs, one output."*

Watch For

Emphasize: this is the first time we are modeling a DECISION, not a physical process. Same tools, different application.

Slide 5: Activity 1: Components

8 min

Say: *"What factors should we weigh? How bad is the problem? How risky is the fix? How much do we know?"*

Watch For

Guide sorting: Three externals (severity, risk, evidence) and one internal (justified). Discuss: Why are these the right factors?

Slide 6: Activity 2: Connections

8 min

Say: *"Severity and evidence PUSH toward action. Risk PUSHES against action. What happens when they fight?"*

Watch For

Draw arrows. Key insight: two positive and one negative arrow on the same output. This is what makes decisions hard.

Slide 7: Activity 3: Simulation

10 min

Say: *"Easy case, easy case, then the HARD case. Predict before we run each one."*

Watch For

Run all three scenarios. Spend extra time on the 'Hard Call' — have students debate what they would do.

Slide 8: Discoveries

5 min

Say: *"Models do not tell us what is right. They tell us what we are WEIGHING. That clarity is the first step to a good decision."*

Watch For

Discuss: Where do you and your classmates disagree? Is it about the severity, the risk, or the evidence?

Slide 9: STEM Challenge

12 min

Say: *"Time to put your reasoning to the test — in a courtroom debate!"*

Watch For

> Assign teams and dilemmas. Give prep time, then run structured debates with jury deliberation.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Problem Severity, Solution Risk, Available Evidence
- **INTERNAL:** Intervention Justified

Relationships

- **Problem Severity to Intervention Justified:** POSITIVE (+) — The more severe the problem, the more justified intervention becomes. A species going extinct demands more action than a minor population dip.
- **Solution Risk to Intervention Justified:** NEGATIVE (-) — The riskier the solution, the LESS justified intervention becomes. A solution that might cause more harm than good is harder to justify.
- **Available Evidence to Intervention Justified:** POSITIVE (+) — The more scientific evidence supporting the solution, the more justified intervention becomes. Acting on strong research is more defensible than acting on a hunch.

Key Discoveries

- Most ethical decisions are NOT clear-cut — they involve competing factors pulling in different directions
- Problem severity, solution risk, and available evidence ALL matter — you cannot ignore any of them
- The 'hard calls' happen when factors conflict — severe problem but risky solution
- Models cannot tell us what is RIGHT, but they can help us see all the factors clearly

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Science alone can answer ethical questions"	Science provides DATA — how severe is the problem, what does the evidence show, what are the risks. But the DECISION of what to do involves values, priorities, and moral principles that science alone cannot determine. Two people can agree on all the facts and still disagree on the right action because they weight the factors differently.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If the technology exists, we should use it"	Just because we CAN do something does not mean we SHOULD. Nuclear weapons exist but most nations agree not to use them. Gene editing can modify human embryos but most countries have banned it. The existence of a tool says nothing about whether using it is wise. Our model helps evaluate each case individually.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Doing nothing is always the safe option"	Inaction is itself a choice with consequences. If an ecosystem is collapsing and we choose not to intervene, we are choosing to let it collapse. Sometimes the risk of doing nothing exceeds the risk of acting. Our model captures this through the Problem Severity factor — when severity is extreme, the cost of inaction is also extreme.